

Statistical mechanics of exploding phase spaces

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Abstract:

Real complex systems, as encountered in biology or neuroscience, typically involve components that interact and create new emergent states leading to phase spaces whose volumes grow super- exponentially (“exploding”) with the number of degrees of freedom. We argue that the standard ensemble theory could break down for such cases and illustrate this phenomenon using simple models. We present a rigorously defined entropy which is extensive in the micro canonical, equal probability, ensemble for super- exponentially growing phase spaces. We suggest that this entropy may be useful in determining probability measures for such systems through appropriately constrained maximum entropy procedures. Work done in collaboration with Roozbeh Pazuki, Gunnar Pruessner and Piergiulio Tempesta